



Uptake of per- and polyfluoroalkyl substances (PFAS) by soybean across two generations

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ABSTRACT

The uptake of per- and polyfluoroalkyl substances (PFAS) by agricultural plants from contaminated soil poses a threat to human health and the ecosystem. The presence of PFAS in plant seeds could also affect their response to PFAS in the environment. In this study, the impact of PFAS exposure on soybean uptake of these compounds across generations was investigated. The results showed that the exposure of the first-generation soybean to a PFAS mixture led to accumulation of perfluorobutanoic acid (PFBA, $12.20 \pm 11.44 \mu\text{g/kg}$), perfluorohexanoic acid (PFHxA, $55.66 \pm 42.06 \mu\text{g/kg}$), and perfluorobutanesulfonic acid (PFBS, $3.24 \pm 2.06 \mu\text{g/kg}$) in the seeds. The progeny soybean plants grown from the first-generation seeds had a higher capability of taking up and accumulating perfluorocarboxylic acids (PFCAs), including PFBA, PFHxA, perfluoroheptanoic acid (PFHpA), and perfluorooctanoic acid (PFOA), when exposed to the same PFAS again. In contrast, the accumulation of perfluorosulfonic acids (PFSAs), including PFBS, perfluorohexane sulfonate (PFHxS), and perfluorooctanesulfonic acid (PFOS), in the second-generation soybean decreased compared to that in the first-generation. As an alternative to PFOA, GenX had a higher plant uptake compared to PFOA in the first-generation soybean, but its uptake by the second generation decreased significantly. Based on findings from this study and considering the widespread nature of PFAS, soybean seeds may need to be quantified regarding PFAS concentrations before their use in agricultural fields. The different uptake of different PFAS, especially increased accumulation of PFCAs by soybean derived from PFAS contaminated seeds, could be further explored for the benefit of phytoremediation.

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) have been extensively used in commercial and industrial products in the past decades and reported to be widely distributed in the environment due to their persistent nature (Buck et al., 2011; Paul et al., 2008; Zhao et al., 2014). Previous studies have demonstrated uptake of PFAS by plants and subsequent entry to the food chain (Costello and Lee, 2020; Ghisi et al., 2019; Lesmeister et al., 2021). There is also countless toxicological evidence in the literature that some PFAS have adverse health effects on plants, animals, and humans (Fenton et al., 2021; Hekster et al., 2003; Lau, 2015).

Soybean has high agro-economic value and diverse uses in both agriculture and industry due to its high contents of protein (~40%) and oil (~20%) (Pagano and Miransari, 2016). The United States Department of Agriculture (USDA) reported that the soybean production in 2020 reached 4.14 billion bushels, making it the second highest valued crop in the U.S. after corn (Grassini et al., 2021). Researchers have reported the uptake and bioaccumulation of PFAS by soybean under different conditions. Wen et al. (2016) found that soybean can take up perfluoro-

rooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS) from biosolids-amended soil and bioaccumulated these PFAS in plant roots and shoots. The concentrations of PFOA and PFOS in soybean roots correlated positively with root protein contents while negatively with root lipid contents. Zhao et al. (2018) reported that perfluorooctane sulfonamide (FOSA), a PFAS precursor, was bioavailable to soybean in a hydroponic system and could be metabolized to PFOS, perfluorohexane sulfonate (PFHxS), and perfluorobutanesulfonic acid (PFBS) during the process of plant uptake. Plant uptake and biotransformation of another PFAS precursor, 8:2 fluorotelomer alcohols (8:2 FTOH), was also observed in soybean (Zhang et al., 2016). While PFAS accumulation in plant shoots and roots has been reported extensively (Costello and Lee, 2020; Ghisi et al., 2019; Jiao et al., 2021; Lesmeister et al., 2021; Wang et al., 2020b), PFAS in seeds were not reported much. A few studies disclosed much lower PFAS concentrations in seeds or grains compared to other tissues of the same plant (Scher et al., 2018; Wen et al., 2014). Regarding soybean, the edible seeds are economically one of the most important food groups in the world, providing vegetable protein and other essential nutrients for humans (Capriotti et al., 2014). PFAS presence in soybean seeds would then raise health concerns due to their persistent and toxic properties. From a practical perspective, seeds are not used if they are known to be contaminated. Given the wide presence of PFAS in various environmental matrices, for instance surface

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water, sediment, soil and even rainfall (Martin et al., 2019; Podder et al., 2021; Rankin et al., 2016; Wang et al., 2016; Wen et al., 2014), however, it could be common that soybean seeds are contaminated by PFAS and these contaminated seeds are not tested for these compounds and planted later. Thus, it is intriguing to see how PFAS-contaminated seeds behave in a PFAS contaminated environment in terms of growth and uptake of these emerging contaminants.

For this purpose, in this study, we grew soybean (*Glycine max* (L.) Merr. cv. Tohya) in soil spiked with PFAS mixtures at a target concentration of 10 or 100 µg/kg, in addition to one control group without PFAS exposure, across two generations. The PFAS mixture consisted of four perfluorocarboxylic acids (PFCAs) (i.e., perfluorobutanoic acid (PFBA), perfluorohexanoic acid (PFHxA), perfluoroheptanoic acid (PFHpA), PFOA), three perfluorosulfonic acids (PFSAs) (i.e., PFBS, PFHxS, PFOS) and a PFOA alternative (i.e., undecafluoro-2-methyl-3-oxahexanoic acid (GenX)). Another two groups of plants were grown in soil spiked with individual PFOA or PFOS at 10 or 100 µg/kg. The concentrations of PFAS in plant tissues at the same growth stage across the two generations were compared. The PFAS concentrations in the seeds of soybean plants exposed to PFAS were determined as well. To our best knowledge, this is the first study reporting how PFAS-contaminated soybean seeds behave differently from the uncontaminated counterparts.

2. Materials and methods

2.1. Soil preparation

The chemicals and reagents used in this study were listed in Table S1. The sandy loam soil (pH = 7.56) used for plant cultivation was collected from a local farm in Albany County, NY. This soil has an organic matter content of 4.82% and total organic carbon content of 2.44%. Before use, the soil was passed through a 2-mm sieve and mixed with vermiculite in a 1:1 volume ratio. The inclusion of vermiculite was to improve soil aeration and water retention. The prepared soil was then allocated to 18-fl oz disposable plastic cups. Each cup contained 170 g of dry soil. Afterwards, the stock solution of PFOA, PFOS, or PFAS mixture containing PFBA, PFHxA, PFHpA, PFOA, PFBS, PFHxS, PFOS, and GenX was added to each cup, reaching the target concentration of 10 µg/kg or 100 µg/kg for each individual PFAS. To homogenize PFAS and soil, water was further added to each cup until the soil reached 100% of field capacity. Controls without PFAS spiking were set up as well. Six replicates were prepared for each treatment and control.

2.2. Plant cultivation and harvest

The seeds of *Glycine max* (L.) Merr. cv. Tohya (soybean) purchased from Johnny's Selected Seeds (Winslow, ME) were placed in wet sand for germination. After 7 days, healthy seedlings were individually transferred to the cups filled with the prepared soil. The plants were then cultivated in a greenhouse and irrigated every day. The temperature and relative humidity in the greenhouse were controlled at 25 – 30°C and 60 – 80%, respectively. Only natural sunlight was provided for plant growth. On Day-41 after germination, just before the flowering stage of the plants, three of the six replicates in each treatment and control were harvested and separated into roots and shoots. The shoots and roots after rinsing by deionized water were freeze-dried at -37°C for 48 h and the soil was air-dried in a fume hood for 14 days. The dried plant tissues and soil were subject to PFAS extraction and quantification. On Day-94 after germination, the remaining three plant replicates in each treatment and control reached full maturity and were harvested. At this stage, all leaves were yellow and fell from the stems. From each mature plant, 10 – 20 fully developed seeds were collected. Half of the collected seeds in each treatment and control were subject to PFAS extraction and quantification and the other half were germinated in wet sand again. After 7 days of germination, healthy seedlings derived from

seeds exposed to PFAS were individually transplanted to the same soil spiked with PFOA, PFOS, or a PFAS mixture at 10 µg/kg or 100 µg/kg for each individual PFAS and cultivated in the same greenhouse. Same as the first-generation, on Day-41 after germination, the second-generation soybean plants were harvested and separated into roots and shoots. Similarly, the freeze-dried plant tissues and air-dried soil were subject to PFAS extraction and quantification.

2.3. PFAS extraction and analysis

PFAS in the freeze-dried soybean shoots/roots, seeds and the air-dried soil were extracted using previously developed methods (Zhang et al., 2021; Zhang et al., 2020; Zhang et al., 2019), which were detailed in the Supporting Information (SI). Each sample was spiked with 10 ng of ¹³C₂-PFHxA as the surrogate for determining the PFAS extraction efficiency. PFAS in the extracts were quantified using a 1290 Infinity II LC system coupled with a 6470 Triple Quad Mass Spectrometer (LC-MS/MS, Agilent Technologies, Santa Clara, CA). The detailed instrumental parameters were listed in the SI. The detection limits of studied PFAS and PFAS recovery in plant tissues and soil were listed in Table S2

2.4. Data analysis

The experimental results are presented as means ± the standard error of three replicates. One-way ANOVA with Duncan's post hoc tests and student t tests were performed with IBM SPSS Statistics 22. Statistical significance was accepted when $p < 0.05$.

3. Results

3.1. PFAS accumulation in soybean seeds

This study used the soybean seeds collected from a previous work where soybean was exposed to PFOA, PFOS or a PFAS mixture at either 10 or 100 µg/kg (Jiang et al., 2022). At the end of that study which was referred to as the first-generation and when soybean reached the end of its reproductive stage on Day-94, we collected the seeds. The harvested seeds were then subject to PFAS extraction and quantification. Exposure to a PFAS mixture at 10 µg/kg or individual PFOA or PFOS at 10 or 100 µg/kg did not lead to any PFAS accumulation in the plant seeds, while exposure to a PFAS mixture at 100 µg/kg resulted in the detection of short chain PFAS including PFBA, PFHxA, and PFBS in the harvested seeds (Fig. 1). PFHxA had the highest concentration of 55.66 ± 42.06 µg/kg, followed by PFBA at 12.20 ± 11.44 µg/kg and PFBS at 3.24 ± 2.06 µg/kg.

3.2. Uptake of PFAS by shoots and roots of soybean grown from PFAS-containing seeds

Without PFAS spiking, all target PFAS in this study were non-detectable in the soil and control plants, indicating that the cultivation environment and the original seeds were PFAS free. As shown in Fig. 2 and 3, after 41 days' exposure to PFAS, all PFAS were detected in the plant tissues. In general, higher PFAS concentrations were observed in soybean shoots and roots when the exposure concentrations were higher. In soybean shoots exposed to the PFAS mixture, short chain PFAS had higher concentrations than long chain counterpart. At the same time after germination (Day-41), we compared the PFAS levels in the plant tissues across two generations. Notably, the concentrations of all PFAS except PFOS in the second-generation plant shoots were significantly higher than those in the first-generation plant shoots. Specifically, PFOA concentration in the second-generation soybean shoots exposed to PFOA only at 10 and 100 µg/kg increased by 9,452.55% and 740.76%, respectively, compared to that in the first-generation soybean shoots. On the contrary PFOS concentration in the second-generation soybean

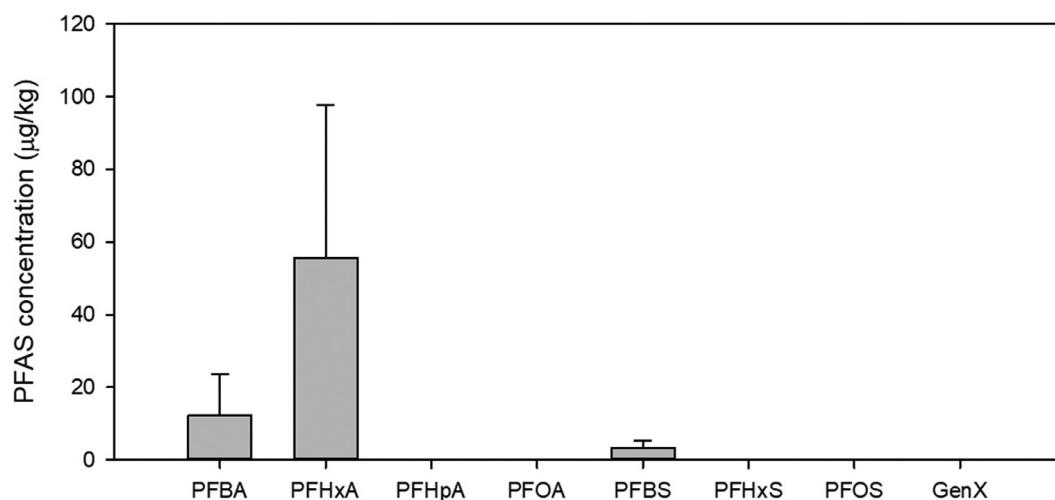


Fig. 1. Concentrations of PFAS in the seeds of the first-generation soybean plants after 94 days of exposure to mixture at 100 µg/kg. Error bars represent standard deviations (n = 3).

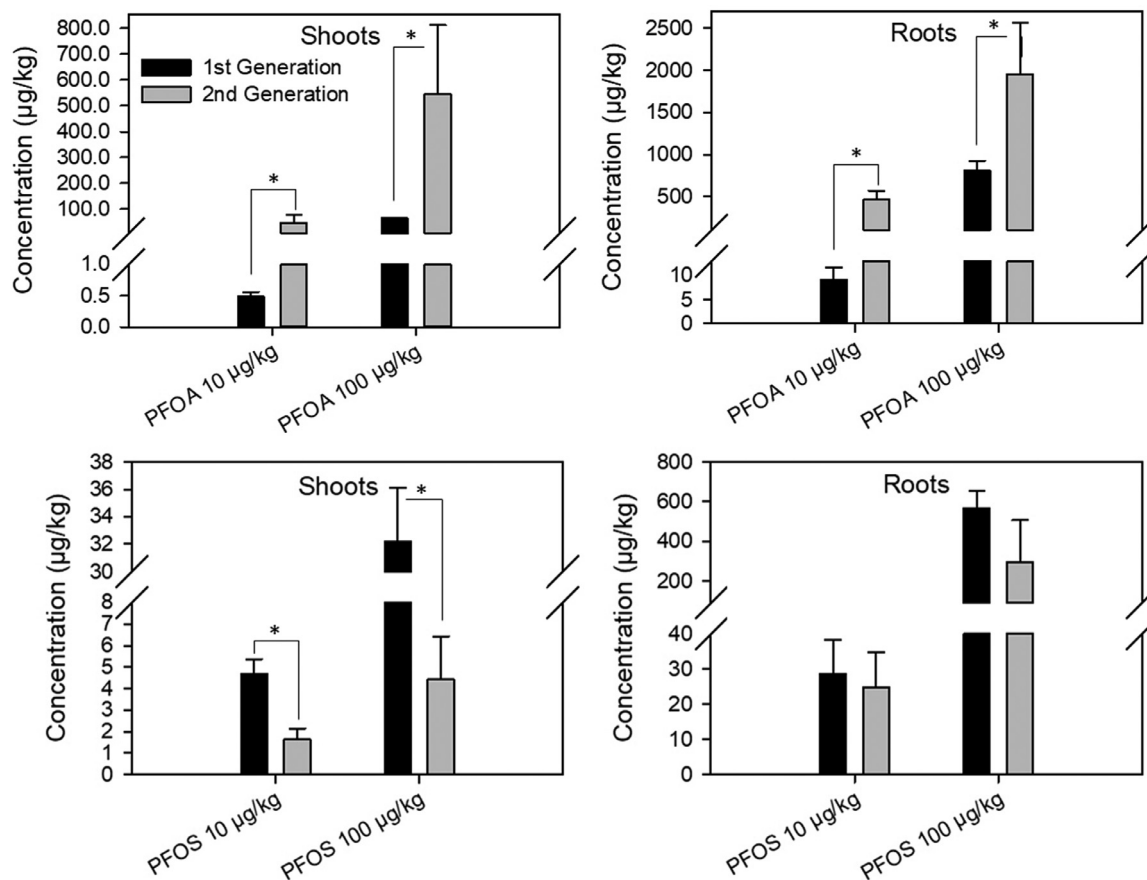


Fig. 2. Concentrations of PFOA or PFOS in the shoots and roots of the first- and second-generation soybean plants after 41 days of exposure to individual PFOA or PFOS at 10 or 100 µg/kg. Error bars represent standard deviations (n = 3). Asterisks indicate significant differences between PFAS concentration in plant tissues exposed to individual spike PFOA/PFOS and PFAS mixture (Student's t-test, $p < 0.05$).

shoots exposed to only PFOS at 10 and 100 µg/kg decreased by 55.77% and 84.83%, respectively (Fig. 2). When exposed to a PFAS mixture at 10 µg/kg, the uptake by the second-generation shoots, when compared to those from the first generation, increased the most for PFBA by 7,446.27%, followed by PFOA (3,858.54%), PFHxA (3,472.94%), and PFHpA (1,387.94%) (Fig. 3). The uptake increase of PFASs in plant shoots across the two generations were relatively lower than those of PF-

CAs. The uptake of PFBS and PFHxS increased by 173.13% and 30.44%, respectively. Furthermore, the second-generation soybean shoots took up 64.09% more of GenX compared to the first-generation plants. At the exposure concentration of 100 µg/kg, a similar trend was observed that the second-generation shoots accumulated more PFAS than the first-generation shoots and the uptake increase of PFCAs across the two generations was higher than those of PFASs.

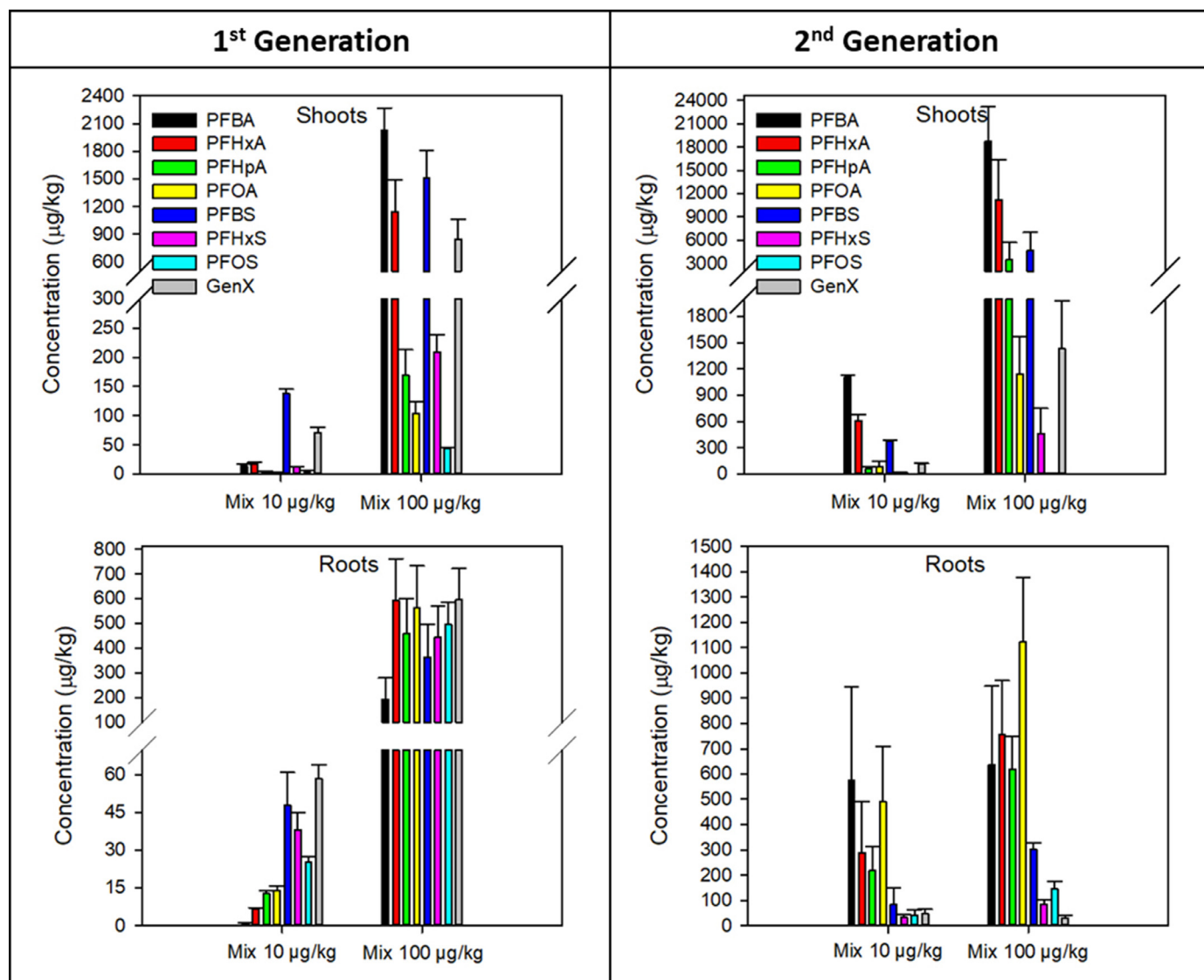


Fig. 3. Concentrations of PFAS in the shoots and roots of the first- and second-generation soybean plants after 41 days of exposure to mixture at 10 or 100 µg/kg. Error bars represent standard deviations (n = 3).

Regarding the PFAS concentrations in plant roots, the second-generation soybean plants exposed to individual PFOA still had significantly higher PFOA concentration than the first-generation plants (Fig. 2). For those exposed to a PFAS mixture, the second-generation soybean roots also had higher concentrations of PFCAs than those in the first-generation roots (Fig. 3). However, at the exposure concentration of 10 µg/kg, PFHxS and GenX in the second-generation soybean roots decreased by 17.18% and 20.61%, respectively, compared to that in the first-generation soybean plants. At the exposure concentration of 100 µg/kg, the uptake of PFBS, PFHxS, and PFOS by the second-generation soybean roots decreased by 17.04%, 80.83%, and 70.45%, respectively. The concentration of GenX in the second-generation soybean roots also decreased by 94.80% compared to that in the first-generation soybean roots.

The root concentration factors (RCFs) of PFAS, which were calculated by dividing the concentration of PFAS in root tissues by their corresponding concentration in the soil, were shown in Fig. 4 and 5. In general, RCFs of PFAS except GenX in the second-generation soybean exposed to PFAS at 10 µg/kg were significantly higher than those in the first-generation soybean. At the exposure concentration of 100 µg/kg, RCFs of PFAS except GenX in two generations of soybean plants

were comparable. RCF of GenX in the second-generation soybean was significantly lower than that in the first-generation soybean, regardless of the exposure concentration. The bioconcentration factors (BCFs) of PFAS were also calculated by dividing the concentration of PFAS in shoot tissues by their corresponding concentration in the soil (Table S3). BCFs of PFAS except PFOS and GenX in the second-generation soybean exposed to PFAS at 10 µg/kg were significantly higher than those in the first-generation soybean. Similarly, at the exposure concentration of 100 µg/kg, BCFs of PFAS except PFOS in the second-generation soybean were significantly higher than those in the first-generation soybean. The translocation factors (TFs) of PFAS, which were the ratio of PFAS concentration in the shoots to that in the roots, decreased with increasing carbon chain length in both generations of soybean plants (Fig. 5). At the exposure concentration of 100 µg/kg, the TFs of PFAS in the second-generation plants were significantly higher than that in the first-generation plants. GenX, which has been widely used as an alternative to PFOA, had higher concentration in plant tissues and higher values of TF compared to PFOA at the same spiked concentration in both generations of the soybean. The exact values and percentage changes of RCFs, BCFs, and TFs in different treatments over two generations were listed in Table S3 and S4.

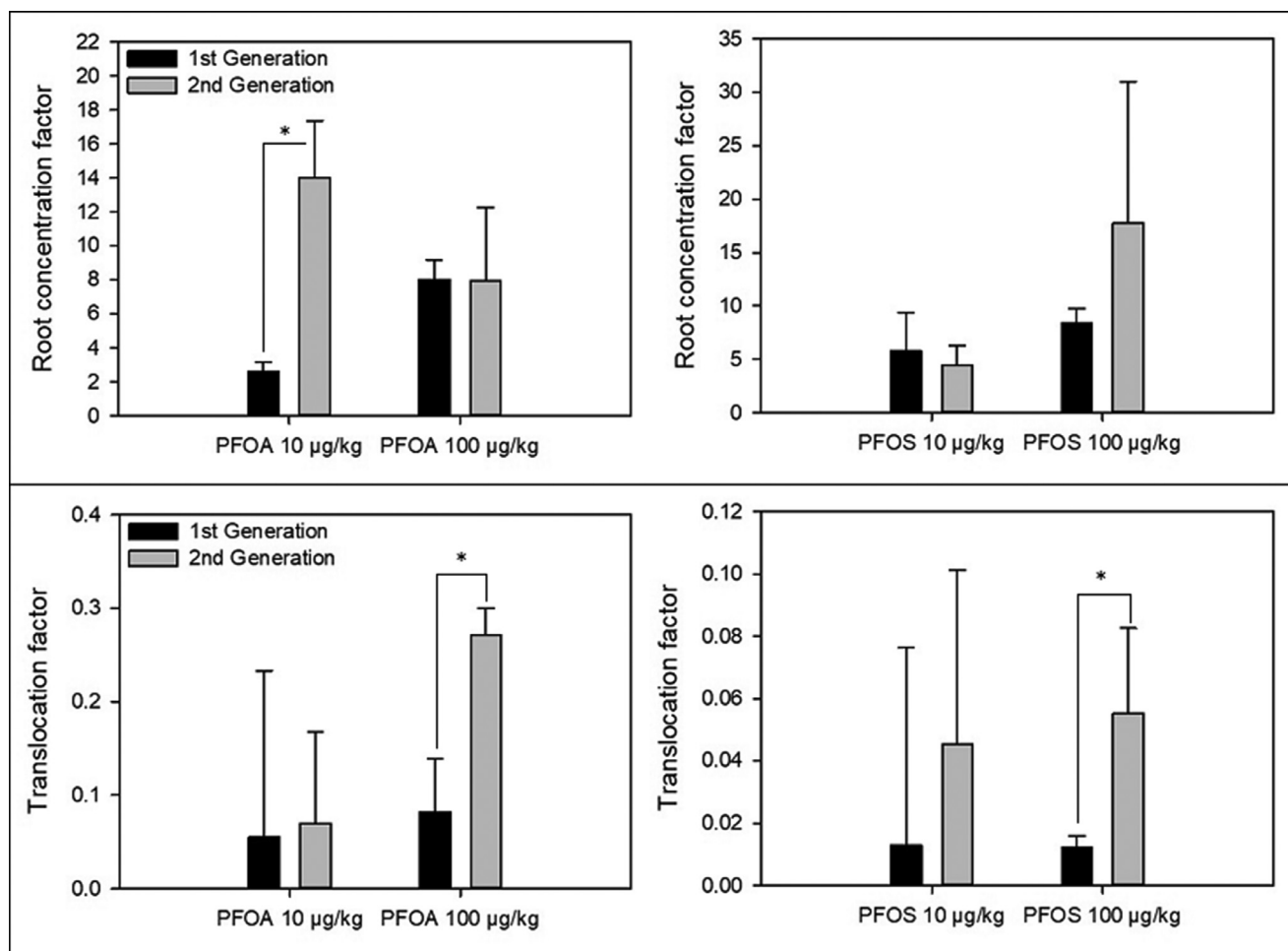


Fig. 4. Root concentration factors (RCF) and translocation factors (TF) of PFOA or PFOS in first- and second-generation plants after 41 days of exposure to individual PFOA or PFOS at 10 µg/kg or 100 µg/kg. Error bars represent standard deviations ($n = 3$). Asterisks indicate significant differences between PFAS concentration in plant tissues exposed to individual spike PFOA/PFOS and PFAS mixture (Student's t -test, $p < 0.05$).

3.3. PFAS removal from soil by soybean

The removal percentages of PFAS on Day-41 after germination, which were calculated by dividing the sum of PFAS mass in plant shoots and roots by the total mass of PFAS added to soil, were compared between the two generations of soybean (Fig. 6). The second-generation soybean plants exposed to individual PFOA at 10 and 100 µg/kg had significantly higher removal than the first-generation, increasing by 1,748.65% and 39.02%, respectively. In contrast, the removal percentages of PFOS by the second-generation exposed to individual PFOS at 10 and 100 µg/kg decreased by 71.13% and 88.77%, respectively, compared to that in the first-generation plants. For the soybean plants exposed to a PFAS mixture, basically, the removal was significantly higher for PFAS with shorter carbon chain (i.e., PFBA, PFHxA, PFBS, GenX) than for PFAS with longer carbon chain (i.e., PFHpA, PFOA, PFHxS, PFOS). This observation was the same for both generations. When comparing the two generations exposed to a PFAS mixture, the second-generation soybean had significantly higher removal of PFCAs than the first-generation, regardless of the exposure concentration. PFBA had the highest increase in removal percentage, from 1.02% and 15.33% to 59.16% and 52.66% across the two generations at the exposure concentration of 10 and 100 µg/kg, respectively. PFHxA was the second. With a PFAS mixture at 10 µg/kg, the second-generation soybean removed 32.39% of this compound. This is significantly higher than 1.34% observed for the first-generation of soybean. Similar observation was also detected for PFBS. The second-generation removed 19.33% vs. 10.68%

calculated for the first-generation. Between the two spiked concentrations of 10 and 100 µg/kg, no statistically significant difference was observed. Removal of the other PFASs and GenX, however, decreased significantly by the second-generation soybean. The exact values and percentage changes of PFAS removal over two generations were listed in Table S3 and S4.

4. Discussion

Information and knowledge regarding effects of multigenerational exposure to emerging contaminants on plants are limited in the literature. Related studies focusing on engineered nanomaterials have reported a wide range of negative effects to different plants as a result of exposure across multiple generations (Ma et al., 2016; Rico et al., 2020; Rico et al., 2017; Singh and Kumar, 2018). With respect to PFAS, however, to the best of our knowledge, there are no available experimental data showing the effects of these compounds to plant over two generations.

By visual observation, the growth and biomass of the two generations of soybean plants did not appear to be affected negatively under different PFAS exposure conditions. But our results did indicate that the exposure of the first-generation plants to PFAS led to the accumulation of short chain PFAS in soybean seeds and changed the PFAS uptake behavior of the second-generation plants. Long chain PFAS were not detected in the first-generation seeds possibly due to their low upward translocation to the shoots. Scher et al. (2018) also reported that irriga-

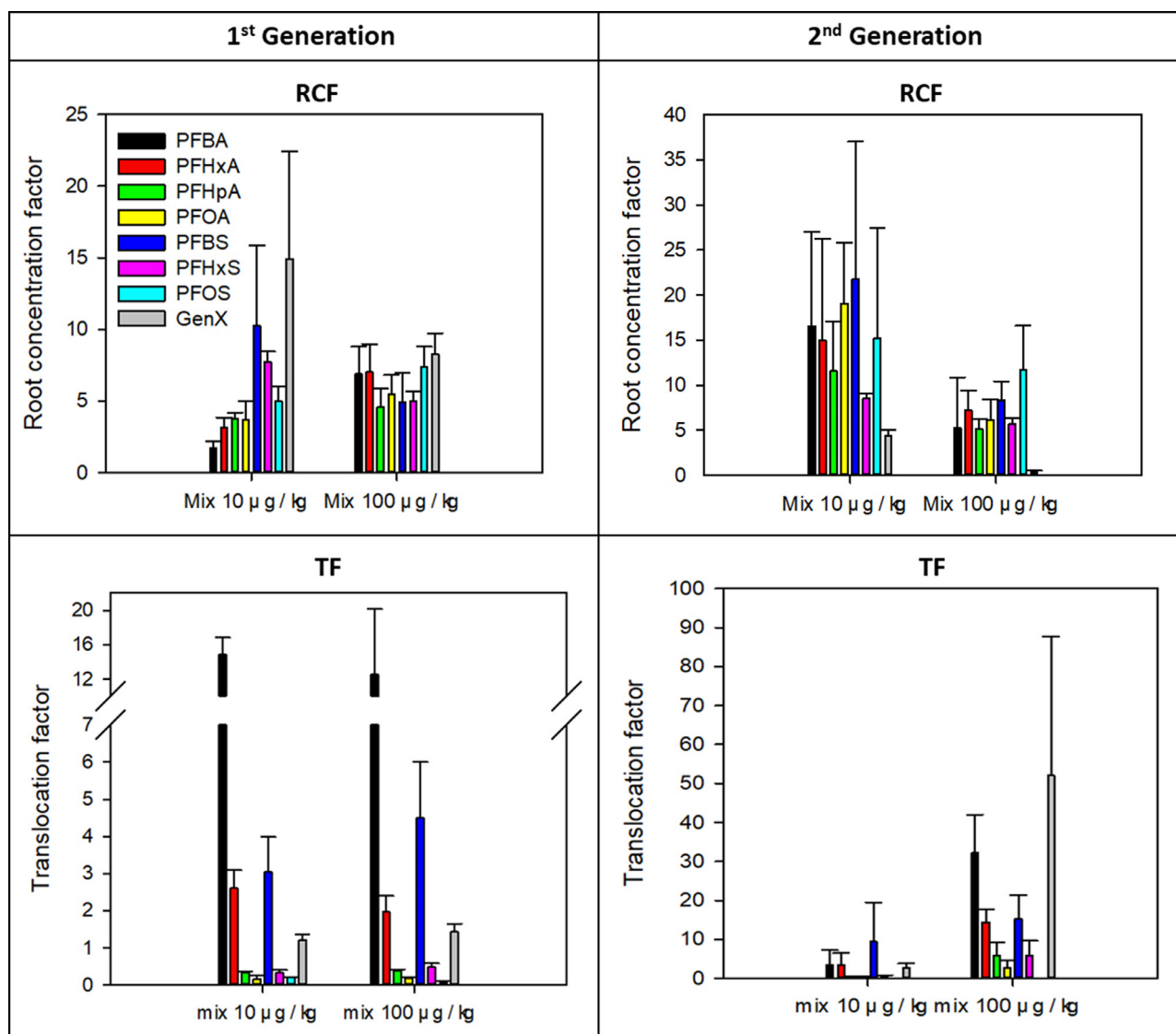


Fig. 5. Root concentration factors (RCF) and translocation factors (TF) of PFAS in first- and second-generation plants after 41 days of exposure to PFAS mixture at 10 µg/kg or 100 µg/kg. Error bars represent standard deviations ($n = 3$).

tion with PFAS (C4 – C8) contaminated water resulted in the accumulation of short chain PFBA at a concentration of up to 20 µg/kg in the seed part of bean, corn, pea, and peapod, demonstrating the accumulation tendency of short chain PFAS in plant seeds. PFBA, PFHxA, and PFBS have shown toxic effects on animals and humans (Klaunig et al., 2015; Liu et al., 2020; Luz et al., 2019; Weatherly et al., 2021). As the main edible part of soybean plants, seeds would eventually transfer the accumulated PFAS to consumers via the food chain, posing risks to the ecosystems.

Apparently, the exposure of the first-generation soybean to PFAS caused significant changes of their uptake by the second-generation. The progeny soybean plants grown from the PFAS-containing seeds showed a significantly higher capability of taking up and accumulating PFCAs (i.e., PFBA, PFHxA, PFHpA, PFOA) and PFBS (at the exposure concentration of 10 µg/kg) when exposed to the same group of PFAS again. As shown in Table 1, there is a positive correlation between what PFAS were in the seeds and what PFAS were removed the most by the second-generation soybean. The significantly enhanced uptake of these PFAS by the plants whose parental generation was affected by PFAS exposure is encouraging from the perspective of phytoremediation. The increased removal percentages of PFCAs and PFBS by the second-generation soy-

bean plants (Fig. 6) would significantly improve the long-term phytoremediation performance at PFAS contaminated sites. Whether this is true or not for other plant species awaits to be verified.

From another perspective, though, the higher accumulation capability of the second-generation soybean for certain PFAS raises serious concerns over food safety. In this study, limited number of seeds harvested from the first generation prevented us from collecting the seeds from the second generation since all triplicates of soybean were harvested before the flowering stage. Thus, the concentrations and the types of PFAS in the second-generation seeds remain unclear at this point. Based on significantly increased PFAS concentrations in the soybean shoots and roots, however, it is reasonable to expect higher concentrations of PFAS in the seeds. Unintentional use of these seeds will inevitably lead to further accumulation of PFAS and other unknown consequences. Therefore, careful evaluations that extend two generations of soybean are necessary to obtain a holistic understanding on the long-term impact of PFAS. Contrary to PFCAs and PFBS, the second-generation soybean plants had a decreased uptake of PFHxS and PFOS compared to the first-generation. The diminished uptake of the two PFASs may imply the development of uptake resistance to this group of PFAS over generations of soybean.

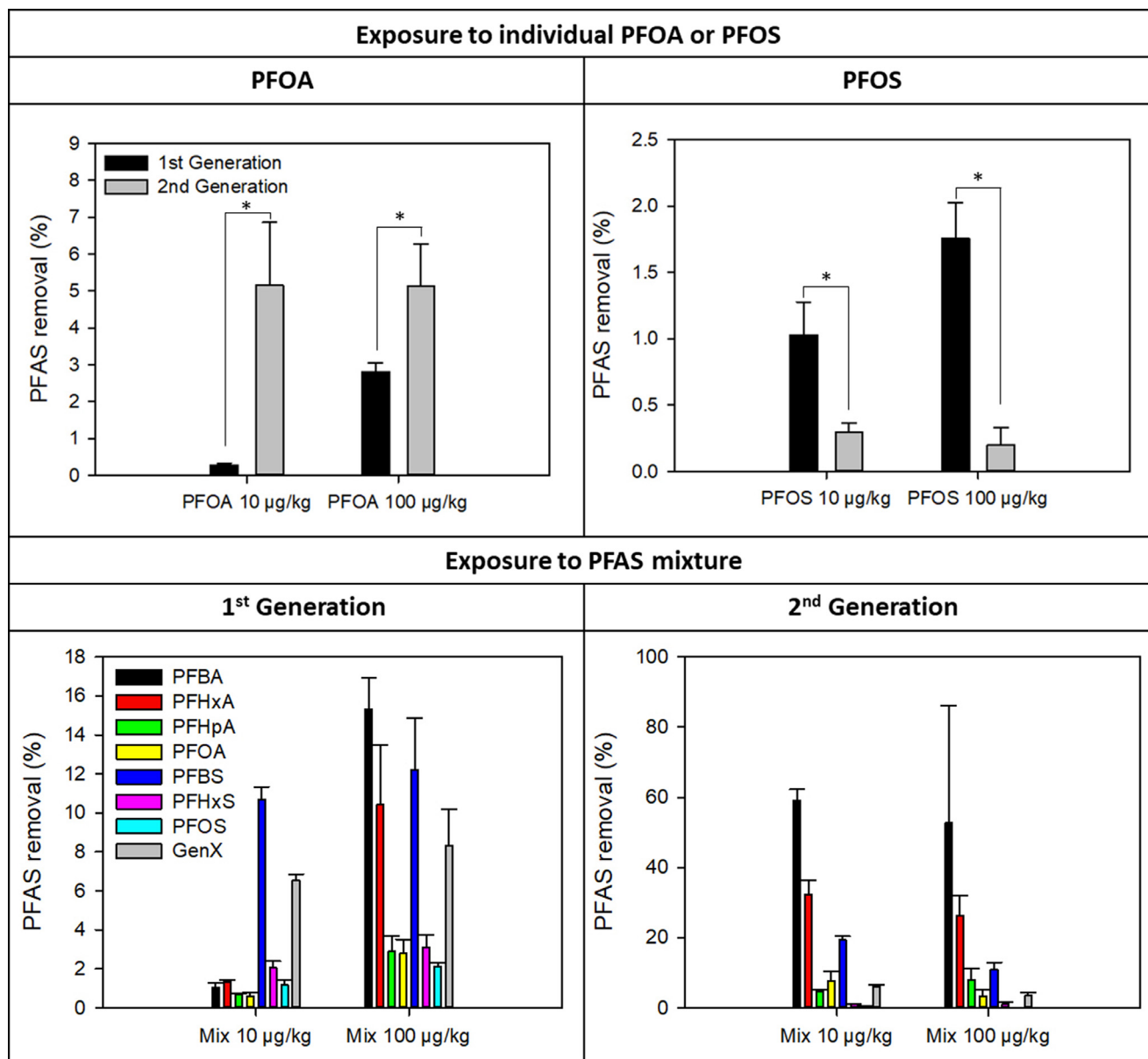


Fig. 6. Percentages of PFAS removal from soil by first- and second-generation soybean plants after 41 days of exposure to individual PFOA, PFOS, or PFAS mixture at 10 µg/kg or 100 µg/kg. Error bars represent standard deviations ($n = 3$). Asterisks indicate significant differences between PFAS concentration in plant tissues exposed to individual spike PFOA/PFOS and PFAS mixture (Student's t -test, $p < 0.05$).

Table. 1

Relationship between PFAS in soybean seeds and PFAS removal by second-generation soybean plants ($n=3$)

Exposure condition: PFAS mixture at 100 µg/kg				
	PFBA	PFHxA	PFBS	PFHpA
PFAS concentration in seeds (µg/kg)	12.20±11.44	55.66±42.06	3.25±2.05	0
PFAS removal by 2nd generation plants (%)	52.66±33.50	26.37±5.67	10.89±2.08	7.93±3.35
	PFOA	PFHxS	PFOS	GenX
PFAS concentration in seeds (µg/kg)	0	0	0	0
PFAS removal by 2nd generation plants (%)	3.34±1.80	1.07±0.44	0.10±0.01	3.44±0.97

Our results indicated that the TFs of PFAS decreased with increasing carbon chain length of PFAS for both generations of soybean, which means short chain PFAS were more readily to be translocated to plant shoots. A previous report showed that the upward translocation of organic contaminants to aboveground tissues is negatively correlated with their hydrophobicity (Miller et al., 2016). Generally, increasing car-

bon chain length of PFAS is associated with increasing hydrophobicity (Gagliano et al., 2020). Thus, short chain PFAS with lower hydrophobicity had higher TFs than long chain PFAS, which is consistent with our observation in this study. Notably, TFs of PFAS except PFOS observed for the second-generation soybean exposed to these compounds at 100 µg/kg were significantly higher than those for the first-generation

soybean, indicating a facilitated translocation pathway for PFAS within plants in the second-generation soybean. Further in-depth studies on plant physiology and biochemistry are required to explore and explain the enhanced translocation of PFAS within the second-generation soybean plants.

Currently, GenX is widely used as an alternative to PFOA in the process of making high-performance polymers for manufacturing cables, cookware, non-stick coatings, etc. (Heidari et al., 2021). Its occurrence in environmental matrices has been reported worldwide (Heydebreck et al., 2015; Wang et al., 2013). Although previous studies have shown the toxicity of GenX to animals and humans (Gaballah et al., 2020; Gebbink and van Leeuwen, 2020), the lifetime health advisory level was finally set at 10 ppt by the U.S. EPA in June of 2022. This level is much higher than 0.004 ppt for PFOA. In the present study, GenX had much lower RCF, but much higher BCF and TF than those of PFOA. Combining these factors together, the removal percentage of GenX was similar to that of PFOA by the second-generation plants at both spiked concentrations (Table 1 and S3). Compared to PFOA, the shorter carbon chain of GenX makes it more hydrophilic (Wang et al., 2019), leading to a higher bioavailability of GenX in soil theoretically. The lower RCF could hint difficulties in passing through the root cell wall and membrane compared to PFOA. The structure of GenX could also contribute to its higher TF in plant tissues. Zhang et al. (2018) reported that the translocation potentials of branched PFOA and PFOS isomers were higher than those of linear isomers due to the higher hydrophilicity of the former. Thus, the branched structure of GenX may also facilitate its transport within soybean tissues.

Compared to the soybean plants exposed to individual PFOA or PFOS, the plants exposed to a PFAS mixture had different PFAS uptake behavior (Fig. 4 and 5). As shown in Table S3, the presence of the PFAS mixture in soil increased RCF, BCF, TF, and removal percentage of PFOA at the exposure concentration of 10 µg/kg for both generations of plants. This signaled enhanced uptake of PFOA by the presence of co-contaminants. At the exposure concentration of 100 µg/kg, the presence of the other PFAS also increased BCF and TF of PFOA but suppressed its RCF. The removal percentage of PFOA was not affected by the co-existence of the other PFAS. This may indicate that the enhancement effect of co-contaminants was offset by the high concentration of PFAS at a total of 800 µg/kg. The lower RCF could hint negative effects of PFAS at high concentrations to the root structures (Liu et al., 2022; Melo et al., 2022). Once PFAS were transferred to the roots, the upward transfer represented by BCF and TF still benefited from the presence of the other PFAS. Regarding PFOS, the presence of other PFAS in the mixture significantly increased its RCF in the second-generation plants when each PFAS was spiked at 10 µg/kg (Table S4). This is reasonable to expect given the fact that PFOS is the most hydrophobic among the target eight and the other PFAS could facilitate its transport from soil to roots (Jiang et al., 2022; Mei et al., 2021). The BCF, TF and removal percentage of PFOS by the second-generation soybean, however, were all less than those observed in the first generation. This is true for both spiked concentrations.

In the environment, PFAS seldomly exist as individual compounds (Brusseau et al., 2020; Podder et al., 2021). The exception could be those contaminated by an industry process utilizing only a limited and known number of PFAS. The presence of PFAS as a mixture in the environment deepens the complexity in remediation efforts. As discussed above, the co-existence of other PFAS could affect an individual PFAS' translocation and distribution in a natural or engineered system. Depending on each PFAS' concentration and physicochemical properties, the effect could be enhancing or suppressing. This brings difficulties to modeling for the purpose of predicting the fate and transfer of PFAS in the environment. Making this matter worse, if the plant seeds are not absolutely PFAS-free, then the use of them can lead to very different results that could be difficult to be explained and compared across different studies even though the same type of seeds are used. All of these point to and demand

high accuracy and reproducibility in PFAS-related research in order to arrive sound conclusions.

In summary, this study revealed differential uptake of PFAS by soybean grown from PFAS-containing seeds. Depending on the specific types of PFAS in the seeds already and each PFAS' physicochemical properties, the translocation of PFAS from soil to roots and then roots to shoots varied significantly. Complete understanding of the observations and results from this study will require detailed investigations of soybean's physiology responding to PFAS exposure. Given the ubiquitous nature of PFAS in the environment and the possible use of PFAS contaminated soybean seeds in agriculture unintentionally, findings of this study are timely and help to raise awareness of potential PFAS in soybean seeds and the multi-generational effects from using such contaminated seeds.

5. Conclusions

This study provided first evidence that soybean grown from uncontaminated and PFAS-contaminated seeds behaved differently toward uptaking the same set of PFAS in the same environment. Our results demonstrated that exposing the first-generation plants to a PFAS mixture at 100 µg/kg led to the accumulation of PFBA, PFHxA, and PFBS in the soybean seeds. The progeny soybean plants grown from the PFAS contaminated seeds had a higher capability of taking up and accumulating PFCAs and PFBS when exposed to the same PFAS again. In contrast to the enhanced uptake of PFCAs and PFBS, the accumulation of PFHxS and PFOS in the second-generation soybean decreased compared to those in the first-generation. GenX had a higher plant uptake compared to PFOA in the first-generation soybean, but its uptake by soybean was less in the second generation.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.hazadv.2022.100170.

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